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MODULE 2

NC PART PROGRAMMING

NC part programming

NC part programming is the procedure by which the sequence of processing steps to be performed on the NC machine is planned and documented

In numerical control, the punched tape must be generated whether part programming is done manually or with the assistance of some computer package.

By the both the method of part programming, punched tape is the net result of programming effort.

Now a days punched tapes are replaced by USB device(Pendrive) and CDs

Part program:

Simply a part program is a computer program to specify

- Which tool should be loaded on the machine spindle?
- What are the cutting conditions (speed, feed, coolant ON/OFF etc.)?
- The start point and end point of a motion segment.
- How to move the tool with respect to the machine?

Simply a part program consists of all information necessary to complete the machining of a component.

Punched Tape

The most common input medium is punched tape.

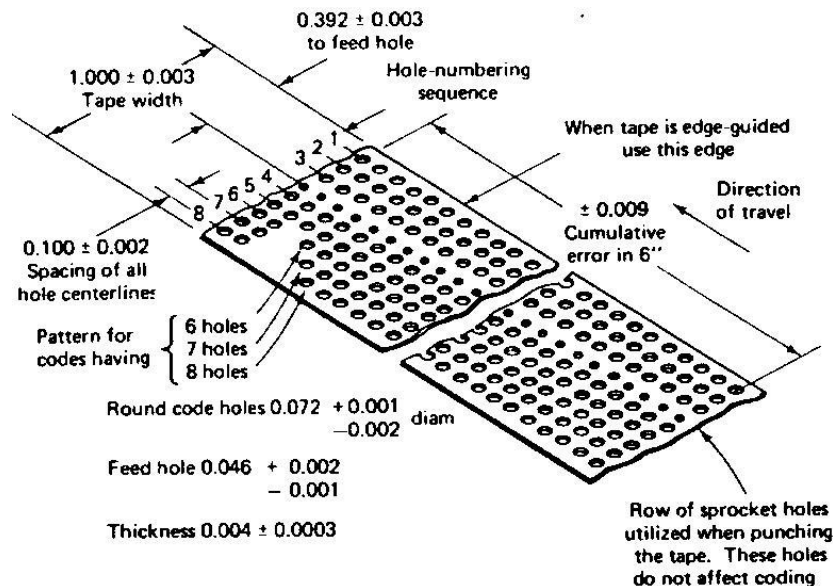


FIGURE 9.1 Numerical control punched tape format as standardized by the Electronics Industries Association (EIA). (Reprinted from Childs [1].)

Fig. The punched tape used for NC is 1 inch wide (25.4 mm) and is standardized by EIA (Electronics Industries Association)

- There are 8 columns of hole in the tape.
- In between 3 and 4 there is one additional column of holes. This is known as sprocket holes which are used to feed the tape through the tape reader.
- The information are stored as binary digits(0 or 1)
- By the row of bits a character (letters, numbers, and symbols)is formed.
- A word is a collection of character.
- Collection of Words form a block
- A block of word forms one set of instruction.
- Each block of information is separated by End-Of-Block symbol(EOB) in the 8th column.

There are 2 methods for preparing punched tape.

- In manual part programming punched tape is prepared with the help of a type writer like device. This produces typed copy of program as well as punched tape

- In Computer Assisted Part Programming (CAPP) tape is prepared directly by the computer using a device called tape punch.
- During production on a conventional NC machine tape is fed through the tape reader once for each work piece.
- It is advanced through the tape reader one instruction at a time
- While the machine tool is performing one instruction, the next instruction is being read into the controllers unit's data buffer.
- After the last instruction has been read into the controller, the tape is rewound back to the start of the program to be ready for the next work part.

PART PROGRAMMING METHODS

1. Manual part programming
2. Computer Assisted Part Programming

STEPS IN MANUAL PART PROGRAMMING

1. Study the component drawing and prepare the most economical sequence of operations in a prescribed format.
2. Choose the NC machine tool and identify the type of material being processed.
3. Decide the work piece datum for each steps.
4. Decide the speed, feed and depth of cut.
5. Decide the tools and type of fixtures to be used.
6. Writing the program in the specified format using NC words.(Manuscript)
7. Input the program and dry run, correct mistakes if any.
8. Produce the first part & inspect and correct the program if any errors in it.
9. Create documentation for future reference. Manual part programming is best suitable for point to point applications.

CONTENTS OF AN NC PROGRAM

Character - row of bits across the tape

Word - sequence of characters (e.g., y-axis position)

Block - collection of words to form one complete instruction

Part program- sequence of instructions (blocks)

NC words

By convention the words in a block are given in the following order

N word- sequence number prefix

G word - preparatory words for positioning, linear and circular interpolation

X, Y, Z words - prefixes for x, y, and z-axes-gives the coordinate positions of the tool

F word- feed rate prefix(F word) specifies the feed in the machining operation

Sword - spindle speed specifies the cutting speed of the process.

T word – Specifies which tool is to be used in the operation. This word would be needed only for machine with a tool operation.

M - miscellaneous command(M word) is used to specify certain miscellaneous or auxiliary functions which may be available on the machine tool. The (M word) is the last word in the block.

EOB-To identify the end of an instruction an End Of Block symbol(EOB) is used.

G-Codes (Preparatory Functions) used in NC part programming

<u>Code</u>	<u>Function</u>
G00	Rapid positioning
G01	Linear interpolation
G02	Circular interpolation clockwise (CW)
G03	Circular interpolation counterclockwise (CCW)
G17	XY plane selection
G18	XZ plane selection
G19	YZ plane selection

G70	Inch input (in.)
G71	Metric input (mm)
G32	Thread cutting
G40	Cutter compensation cancel
G41	Cutter radius compensation left
G42	Cutter radius compensation right
G43	Tool length compensation positive (+) direction
G44	Tool length compensation minus (-) direction
G49	Tool length compensation cancels
G63	Tapping
G84 to G89	Reserved for fixed cycle(Canned cycle)
G90	Absolute programming
G91	Incremental programming
G94	Feed rate mm/minute
G95	Feed rate in mm/rev.

Note : On some machines and controls, some may be differ

M-Codes (Miscellaneous Functions) used in NC part programming

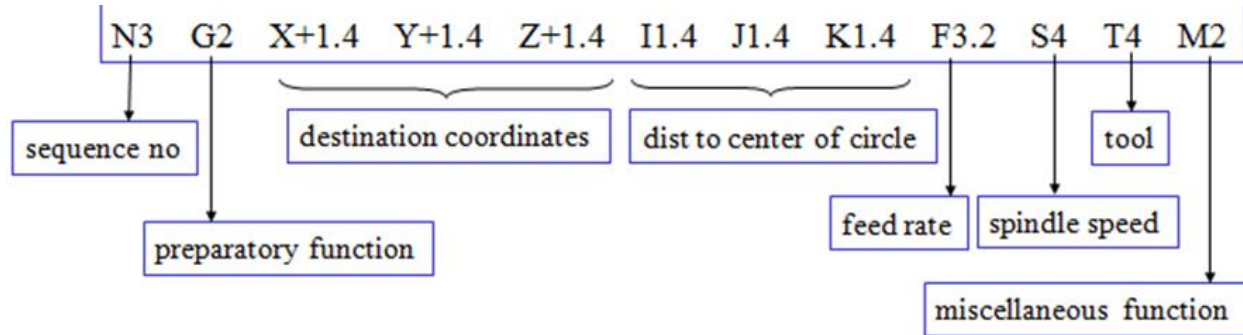
<u>Code</u>	<u>Function</u>
M00	Program stop (unconditional)
M01	Program stop (conditional)
M02	End of program with return to program start
M03	Spindle start (forward CW)
M04	Spindle start (reverse CCW)

M05	Spindle stop
M06	Tool change
M08	Coolant on
M09	Coolant off
M10	Chuck - clamping
M11	Chuck - unclamping
M12	Tailstock spindle out
M13	Spindle start + coolant ON
M15	Spindle OFF+ coolant OFF
M17	End of Subroutine
M30	End of part program

Standard formats in NC part programming

1. Word address format

- In this format alphabets are called addresses.
- The alphabets N,G,X,Y,Z,S,F,T and M are separate addresses giving standard meaning.
- MCU uses these alphabets for addressing a memory location on it
- In this format block of instruction may be of any length and the word can be placed in any sequence since the letter address will search and identify the corresponding word.

Example**2. Tab sequential format**

- In this format words are given in a sequential order.
- For example `N010 G82 X10 Y15 F10 S500 T01 D2 M09 <EOB>` this block of information is given as
`>010 >82 >10 >15 >10 >500 >01 >2 >09 EOB`
- The MCU reads first tab and stores the data in the address corresponding to the sequence number. Then the second word is recognized as preparatory function. Similarly all the word are read and stored in particular addresses.
- If next block contains same except for feed and speed values the format is given as follows

`> > > > >15 >400 > > >`

3. Fixed block format

- In this format the instruction is given in a standard sequence and the block contains a fixed number of characters.
- There are no letter addresses or tab codes
- No words are omitted.
- Even if any data remains the same as in the previous block ,it should be repeated in the next block also

Example

`010 82 10 15 10 500 01 2 09 <EOB>`

- Different CNC machines have different fixed formats.
- For example a CNC lathe has only X and Z word and it does not have any Y word.

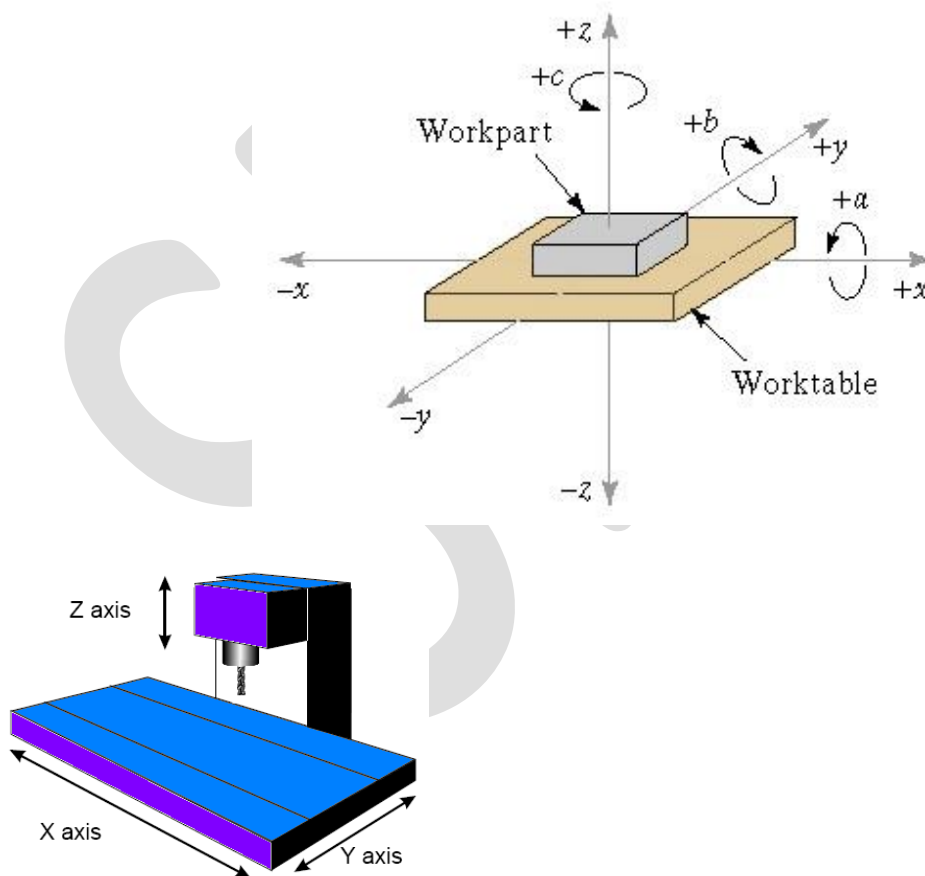
NC CO ORDINATE SYSTEMS

The purpose of coordinate system is to provide a means of locating the tool in relation to the work piece.

For the part programmer to plan the sequence of positions and movements of the cutting tool relative to the workpiece, it is necessary to establish the standard axis system by which the relative positions can be specified.

For flat and prismatic (block-like) parts:

Example Milling and drilling operations, the coordinate system is as follows



For rotational parts

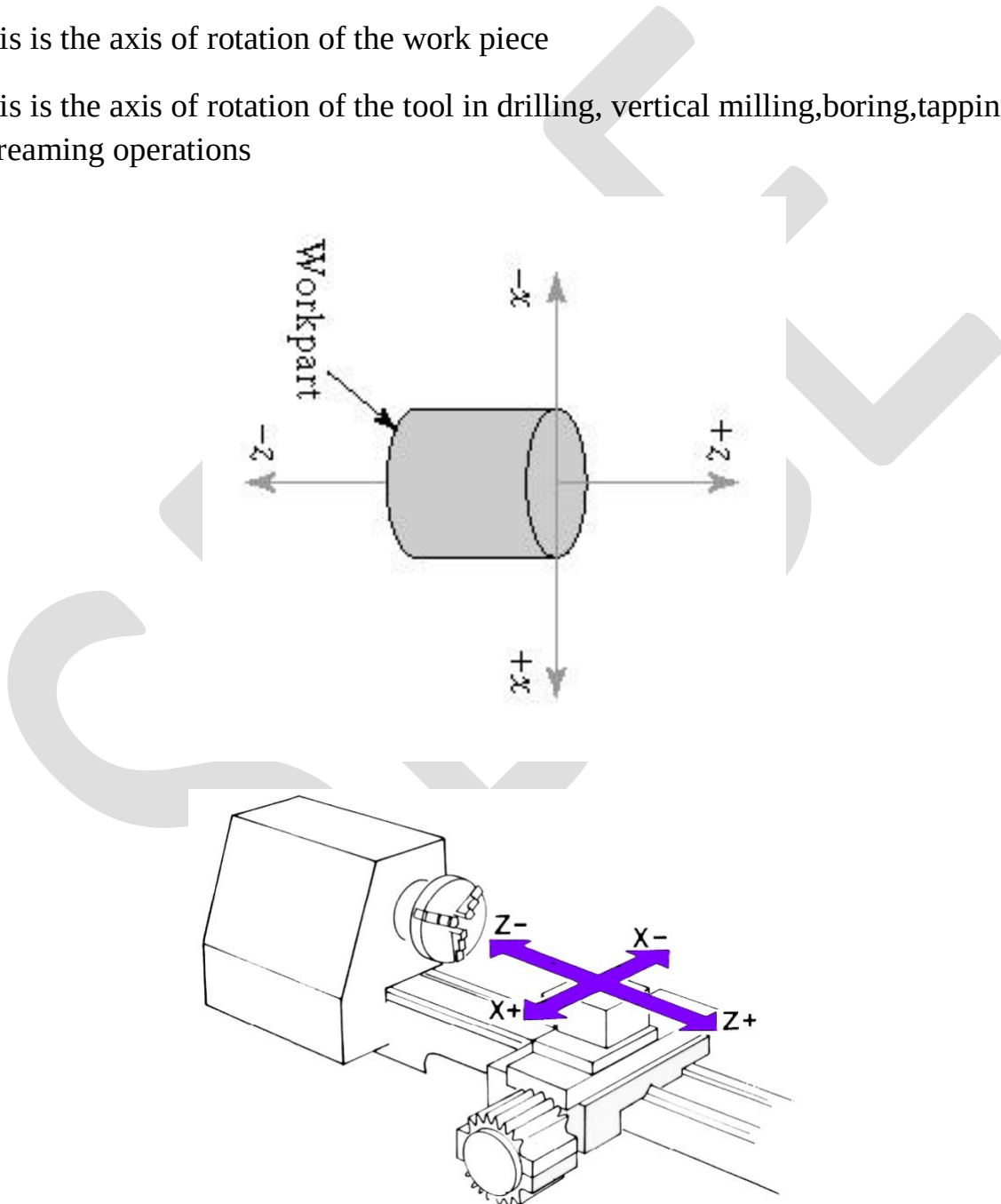
Turning operations

Only x- and z-axes

ie. In CNC lathe X - axis is parallel to cross slide

Z axis is the axis of rotation of the work piece

Z axis is the axis of rotation of the tool in drilling, vertical milling, boring, tapping and reaming operations



To distinguish positive from negative angular motions right hand thumb rule can be applied.

Positive (Clock-wise) of Spindle

Curl right-hand with thumb pointing out in +veX, Y or Z axis direction

The curl of the finger represents the +ve rotation about each axis.

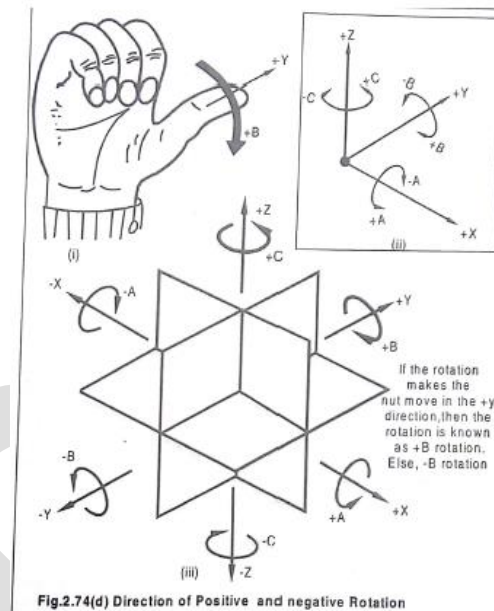


Fig.2.74(d) Direction of Positive and negative Rotation

Fixed zero and floating zero

The programmer must determine the position of the tool relative to the origin (zero point) of the coordinate system.

NC machines have 2 methods for specifying this origin

1. Fixed zero

In this case origin is always located at the same position on the machine table and is usually the south west corner (lower left hand corner) of the table and all tool locations will be defined by positive x & y co ordinates.

2. Floating zero

Machine operator set the zero point at any position on the machine table. The decision is based on part programming convenience.

COMPUTER ASSISTED PART PROGRAMMING (CAPP)

Part programming is done with the assistance of a high speed digital computer.

It is applied in more complicated point to point jobs and in contouring applications.

For this different part programming language has been developed to perform the computations automatically to prepare the tape.

Machining instructions are written like statements of the NC programming language.

Then it is processed by the computer.

CAPP -Explanation

A.Part programmers job in CAPP

1. Defining the work part geometry

Enumerate the elements out of which the part is composed

Each geometric elements must be identified and the dimensions and locations of elements are clearly defined.

For this different geometric statements are used in the program.

2. Specifying the operation sequence

3. Specifying the tool path

The part programmer can use various motion commands to direct the tool to machine along the work part surface, to go to point locations in drilling.

B. Computers job in CAPP

1. Input translation

The part programmer enters the program written in the APT or other language.

The input translation component converts the coded instructions contained in the program into computer usable form preparatory to further processing.

The arithmetic calculation unit of the system consists of a comprehensive set of instructions for solving the mathematics required to generate the part surface.

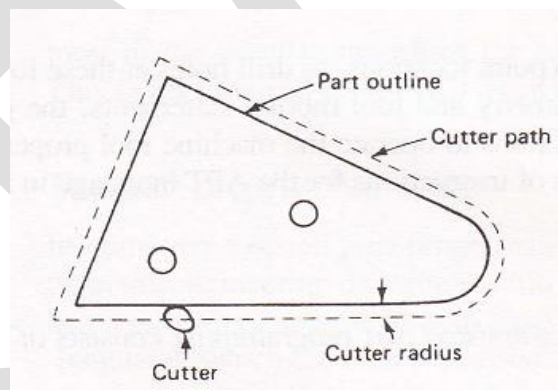
These instructions are called by the various part programming language statements

2. Cutter offset computation

The actual tool path is different from the part outline because the tool path is defined as the path taken by the center of the cutter.

The purpose of cutter offset computation is to offset the tool path from the desired work part surface by the radius of the cutter.

Therefore part programmer can define the exact part outline in the geometry statements.



Cutter offset

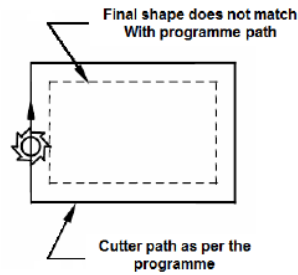


Figure 4.9 : Tool Path without Cutter Compensation

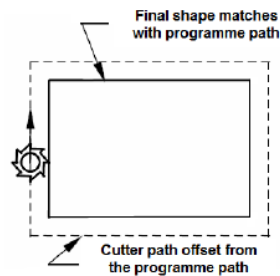


Figure 4.10 : Tool Path with Cutter Compensation

3. Post processor

NC machine tool systems are different and they use different NC tape format.

Nearly all part programming languages, including APT are designed to be general purpose language, not limited to one or two machine tool types.

Therefore in the post processing stage, take the general instructions and make them specific to a particular machine tool system. The unit perform this task is called post processor.

Simply post processor is a separate computer program that has been written to prepare the punched tape for a specific machine tool.

The input to post processor is the out from the other 3 components such as translator, arithmetic calculator, cutter offset data.

The output of the post processor is the NC tape written in the correct format for the machine on which is to be used.

CNC LANGUAGES

The different CNC languages are as follows

1. APT(Automatically Programmed Tools)

2. ADAPT(ADaptation Of APT)
3. EXAPT(EXtended subset of APT)
4. UNIAPT(UNIted APT)
5. SPLIT(Sundstrand Processing Language Internally Translated)

APT (Automatically Programmed Tool) language

- APT is not only an NC language, but also a computer program that performs the calculations to generate the cutter positions based on different NC statements.
- Developed by MIT,US in 1959
- APT is a 3 dimensional system that can be used to control up to 5 axes. Also it is used to control variety of different machining operations.
- There are over 400 words in the APT vocabulary

4 types of NC statements in APT language (Very important)

1. Geometry statements
2. Motion statements
3. Post processor statements
4. Auxiliary statements

1.Geometry statements

Defines the Geometric elements that comprise the work part.

The general form of an APT geometry statement.

Symbol = geometry type /description data

Symbol: used for the identification of a geometric element.

A symbol can be any combination of 6 or fewer alphabetic and numeric characters. At least one of the six must be an alphabetic character.

Geometric type identifies the type geometric element

Geometry type examples

POINT, LINE, CIRCLE, PLANE etc

Descriptive data include quantitative dimensional and positional data, previously defined geometric element and other APT words.

Comas are used to separate the words and numbers in the descriptive data

DESCRIPTIVE DATA

PARLEL, PERPTO, RIGHT, LEFT, INTOF, CENTER, ATANGL, RADIUS etc

Examples

- L3= LINE/P3,P4
- L4 = LINE/P5,PARLEL ,L3
- PL1= PLANE/P1,P4,P5
- PL2= PLANE/P2,PARLEL,PL1
- C1= CIRCLE/CENTER,P1,RADIUS, 5.0

Ground rules that must be followed in formulating an APT geometry statement

- 1. The coordinate data must be specified in the order x,y,z**

P1= POINT/5.0,4.0,0.0

- 2. Any symbol used as descriptive data must have been previously defined.**

P2= POINT/INTOF ,L1,L2

L1,L2 Should be previously defined

- 3. A symbol can be used to define only one geometric element**

P1= POINT/1.0,1.0,1.0

P1= POINT/2.0,3.0,4.0 This is incorrect

Only one symbol can be used to define any given element

P1= POINT/1.0,1.0,1.0

P2= POINT/1.0,1.0,1.0 This is incorrect

- 4. Lines defined in APT are considered to be of infinite length in both directions**

5. Similarly, planes extend indefinitely and circles defined in APT are complete circles.

2.Motion statements:

Used to describe the path taken by the cutting tool

General form is motion command/descriptive data

Example **GOTO/P1**

motion command: which tells the tool what to do

descriptive data: which tell the tool where to go

FROM/TARG

FROM is an APT word which indicates the initial point from which others will be referenced.

TARG symbol given to the starting point

FROM/-2.0,-2.0,0.0

Motion statements are different for PTP and Contouring motions

For PTP motions the commands are GOTO,GODLTA.

GOTO/P2

GOTO/2.0,7.0,0.0

GODLTA command specifies the incremental move for the tool

GODLTA/2.0,7.0,0.0

The GODLTA command is useful in drilling and related operations.

GOTO/P2

GODLTA /0,0,-1.5

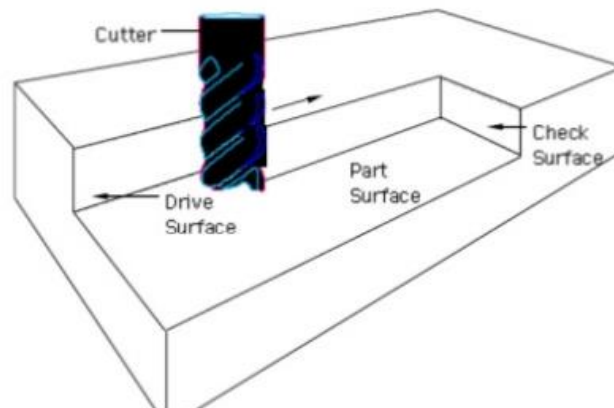
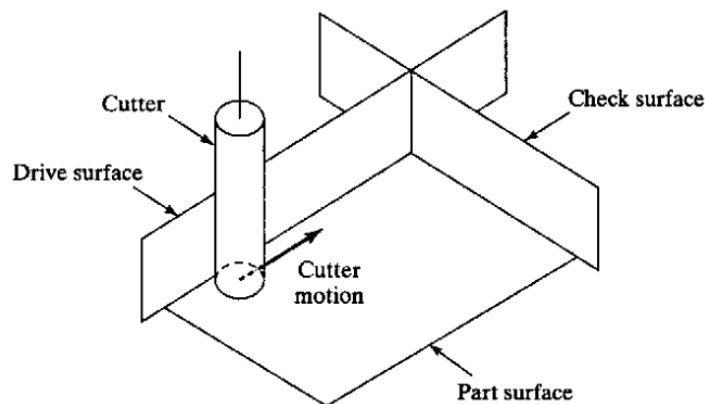
GODLTA/0,0,+1.5

For contouring motions

Drive surface: that guides the sides of the cutter.

Part surface: this is the surface on which bottom of the cutter rides. Part surface may or may not be an actual surface of the work part.

Check surface: This is the surface that stops the movement of the tool in its current direction.

**Modifier words**

There are several ways in which the check surface can be used. This is determined by APT modifier words

TO, ON, PAST, TANTO

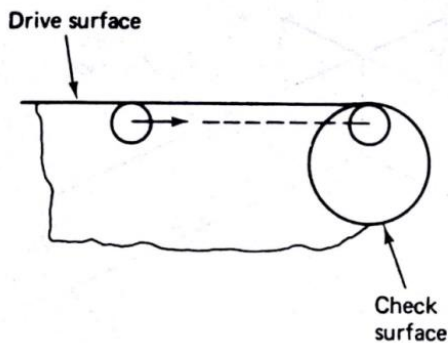
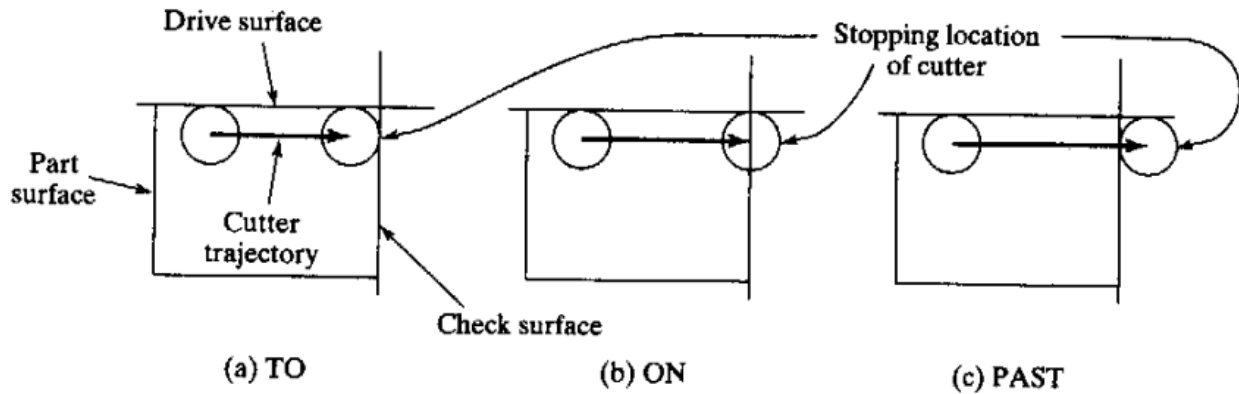


FIGURE 8.12 Use of APT modifier word TANTO. TANTO moves tool to point of tangency between two surfaces, at least one of which is circular.

From statement defines the initial drive surface, part surface and check surface.

The sequence of the following form

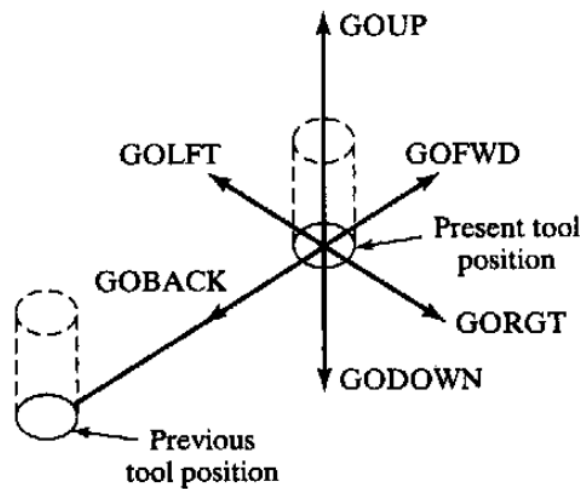
FROM/TARG

GO/TO, PL1, TO, PL2, TO, PL3

GO command instructs the tool to move to the intersection of the drive surface (PL1), part surface (PL2) and check surface (PL3)

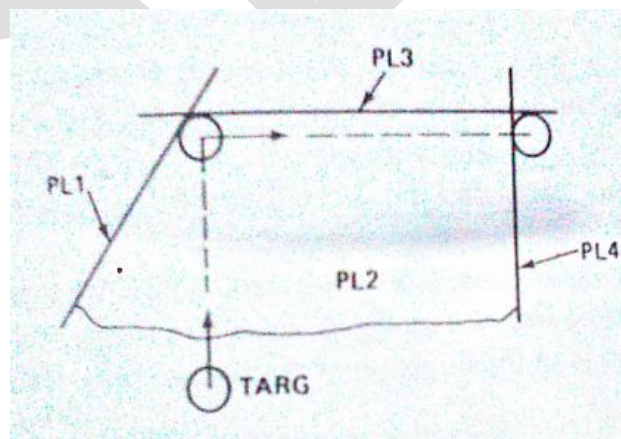
GO TO is used for PTP motions.

GO/TO is used to initialize the sequence of contouring motions.



APT contouring motion commands

- GOLFT commands the tool to make a left turn relative to the last move.
- GORGT commands the tool to make a right turn relative to the last move.
- GOFWD commands the tool to move forward relative to the last move.
- GOBACK commands the tool to reverse direction relative to the last move.
- GOUP commands the tool to move upward relative to the last move.
- GODOWN commands the tool to move down relative to the last move.



The above contouring motions can be explained as follows

FROM/TARG

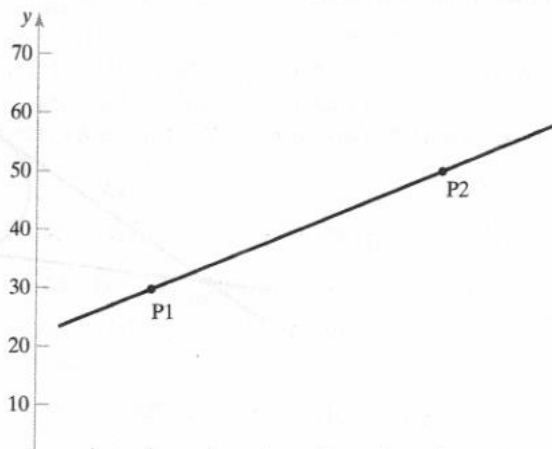
GO/TO, PL1,TO,PL2,TO ,PL3

GORGT/PL3, PAST PL4

Examples to write APT statements for different geometry cases are given below

L1 = LINE/P1, P2

L1 = LINE/20, 30, 0, 70, 50, 0



L1 = LINE/P1, LEFT, TANTO, C1

L2 = LINE/P1, RIGHT, TANTO, C1

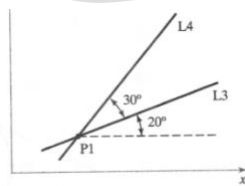
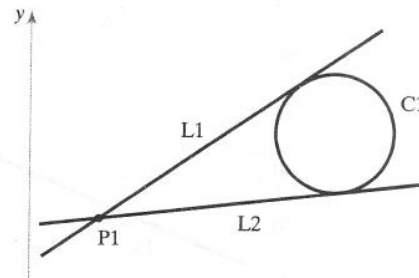


Figure B7.5 Defining a line using a point and the x-axis or another line.

L3 = LINE/P1, ATANGL, 20, XAXIS

L4 = LINE/P1, ATANGL, 30, L3

C1=CIRCLE/CENTER, 4.0, 3.0, 0.0, RADIUS, 2
C1=CIRCLE/4.0, 3.0, 0.0, 2.0

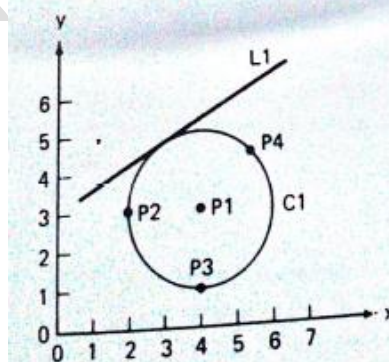


FIGURE A8.1

C1=CIRCLE/CENTER, P1, RADIUS, 2.0

3. By the center point and tangent to a line. See Figure A8.1.

C1=CIRCLE/CENTER, P1, TANTO, L1

4. By three points on the circumference. See Figure A8.1.

C1=CIRCLE/P1, P3, P4

5. By two intersecting lines and the radius. See Figure A8.2.

C2=CIRCLE/XSMALL, L2, YSMALL, L3, RADIUS, .375

C3=CIRCLE/YLARGE, L2, YLARGE, L3, RADIUS, .375

C4=CIRCLE/XLARGE, L2, YLARGE, L3, RADIUS, .375

C5=CIRCLE/YSMALL, L2, YSMALL, L3, RADIUS, .375

CLPRNT: Cutter location print (auxiliary statement). Can be used to obtain a computer printout of the cutter location sequence on the NC tape.

COOLNT: Coolant (postprocessor statement). Turns coolant on, off, and actuates the coolant options that may be available. Examples:

COOLNT/ON	COOLNT/OFF
COOLNT/FLOOD	COOLNT/MIST

CUTTER: Cutter (auxiliary statement). Defines cutter diameter to be used in tool offset computations. The statement

CUTTER/1.0

defines a 1.0-in.-diameter milling cutter. Cutter path would be offset from part profile by one-half the diameter.

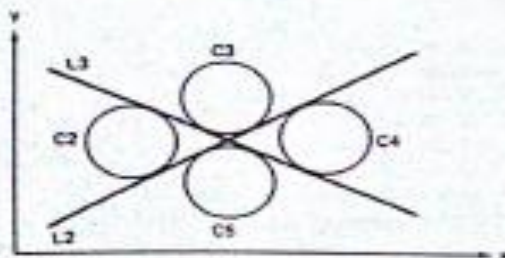


FIGURE A8.2

INTOL: Inside tolerance (auxiliary statement). Indicates the allowable tolerance between the inside of a curved surface and any straight-line segments used to approximate the curve. See Figure A8.3.



INTOL/.005

FIGURE A8.3

POINT: Point (geometry type). Used to define a point.

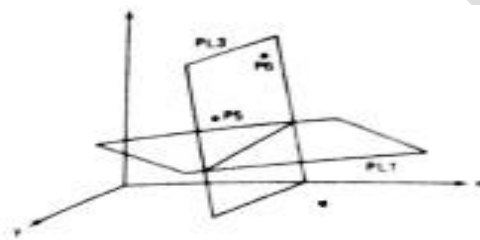


FIGURE A8.14

POINT: Point (geometry type). Used to define a point.
Methods of definition:

1. By the x, y, and z coordinates. See Figure A8.15.
P1=POINT/3.0, 1.5, 0.0

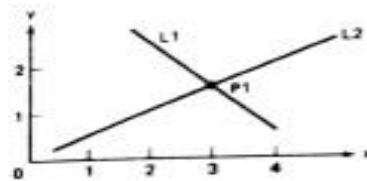


FIGURE A8.15

2. By the intersection of two lines. See Figure A8.15.

P1=POINT/L1, L2

3. By the intersection of a line and a circle. See Figure A8.16.

P2=POINT/YLARGE, INTOF, L3, C1

P3=POINT/XLARGE, INTOF, L3, C1

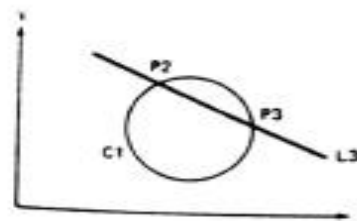


FIGURE A8.16

1. By three points that do not lie on the same straight line. See Figure A8.12.
 $PL1 = \text{PLANE}/P1, P2, P3$

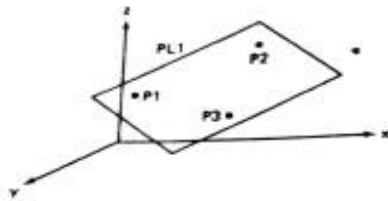


FIGURE A8.12

2. By a point and being parallel to another plane. See Figure A8.13.

$PL2 = \text{PLANE}/P4, \text{PARLEL}, PL1$

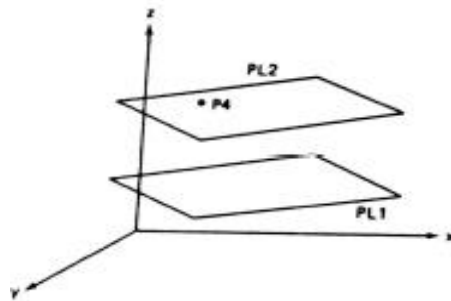


FIGURE A8.13

3. By two points and being perpendicular to another plane. See Figure A8.14.

$PL3 = \text{PLANE}, \text{PERPTO}, PL1, P5, P6$

4. By two intersecting circles. See Figure A8.17.

$P4 = \text{POINT}/\text{YLARGE}, \text{INTOF}, C1, C2$

$P5 = \text{POINT}/\text{YSMALL}, \text{INTOF}, C1, C2$

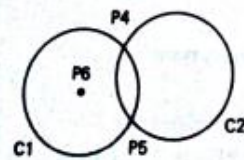


FIGURE A8.17

5. By the center of a circle

$P6 = \text{POINT}/\text{CENTER}, C1$

3. Post processor statements

To write complete part program statements must be written that control the operation of spindle, the feed and other features of the machine tool.

COOLNT/ RAPID

FEDRAT / END

MACHIN/

TURRET/

SPINDL/

Examples

- MACHIN/ : use to specify the machine tool
e.g.- MACHIN/DRILL,3
- COOLNT/ : coolant fluid to be turned on or off.
e.g.- COOLNT /ON
- FEDRAT/: feed rate for the tool in inch per min.
e.g.- FEDRAT/4.5
- SPINDL/: spindle rotation speed in rev per min.
e.g.- SPINDL/ 850

COOLNT/FLOOD turns fluid ON

COOLNT/MIST) means (COOLNT/OFF)

4. Auxiliary statements

The complete APT program must also contain various other statements. These are used for cutter size definition, part identification, cutter location print data and so on,

Auxiliary statements are

PARTNO

CLPRNT

CUTTER

INTOL/

OUTTOL/

FINI

The post processor and auxiliary statements are of 2 types

1. With slash. Require descriptive data after slash
2. Without slash. No additional data are required, self-contained

MACRO STATEMENT IN APT

A very powerful feature of APT is MACRO statement.

It would be used where certain motion sequence would be repeated several times within a program.

The purpose of using a macro statement is to reduce the total number of statements required in the APT program ,thus making the part programmers job easier and less time consuming.

MACRO statement format

Symbol = MACRO/parameter definition(s)

Symbol must be six characters or fewer and at least one of them must be a letter or alphabet.

To activate the MACRO, a **CALL** statement is used.

CALL/ SYMBOL, Parameter specification

Symbol is the name of macro to be called

Parameter specification identifies the particular values of the parameter that are to be used in the execution of macro subroutine.

Drawbacks of NC

1. Part programming mistakes

Syntax errors and lack best sequence of processing steps.

2. Non optimal feeds and speeds

The control system does not provide the opportunity to make the changes in feeds and speeds during the cutting process.

3. Punched tape

Paper tape is fragile and it is subjected to wear and tear .

It is not reliable.

4. Tape reader

Least reliable hardware component of the NC machine. When a breakdown is encountered usually the problem is in the tape reader.

5. Controller

NC controller unit is hard wired. Its control features cannot be easily altered to improve the unit. System is less flexible

6. Management information

Not equipped to provide timely information on operational performance.

Advantages of CNC over conventional NC

1. The part program tape and tape reader are used only once to enter the program into the computer memory. This results improved reliability .Since tape reader is considered as the least reliable component in conventional NC.
2. Increased Program storage capability at the machine tool
3. Tape editing on the machine site. NC tape can be corrected and even optimized.(tool path, speed and feed).
4. Metric conversion: CNC can accommodate conversion of tapes prepared in units of inches into the international system of units.

5. Greater flexibility: This flexibility provides the opportunity to introduce new control options(new interpolation schemes)at low cost .
6. User written programs: for some special applications.
7. Total manufacturing system

CNC is more compatible with the use of a computerized factory wide manufacturing system.

Basic elements of CNC machines.**1. Machine spindle****2. Drives**

Spindle drive : provide drive to the machine spindle.

Feed drive :to drive other axis as per the programme.

For spindle drive large power DC motors are used. Because its speed can be varied infinitely.

For feed drive : DC servomotors ,stepper motors are used.

3. DC servomotors

Constant torque output

Can be start and stop quickly and accurately

Light weight

4. Stepper motors

Rotates in a fixed angular increment

Stepper motors are usually used in open loop control system but an encoder can be used to confirm positional accuracy(200-25000 step/rev).

5. Actuation system

Re circulating ball screw

Nut is replaced by series of balls which circulate in the channel in the form of threads .

The advantages of recirculating ball screws over Acme screws are:

longer life

less frictional resistance

lower torque required

more precise positioning of slides because backlash is almost completely eliminated

This type of leadscrew has an efficiency of up to 90%.

MODULE 2

NC MACHINES (NUMERICAL CONTROLLED MACHINES)

What is Numerical Control (NC) ?

Numerical control can be defined as a form of programmable automation in which the processing equipment (i.e. machine tools) is controlled by coded instructions using numbers, letters and symbols.

Numbers form a set of instructions (NC program) designed for a particular part.

Most important function of NC system is positioning (tool and/or work)

NC technology has been applied in the following areas such as drafting, assembly, inspection, sheet metal press working, spot welding and mainly in machining applications.

Advantages of NC over conventional systems

1. Flexibility with accuracy
2. Repeatability
3. Reduced scrap
4. High production rate
5. Good quality
6. Reduced tooling cost
7. Easy machine adjustments
8. More operations per setup
9. Less lead time
10. Rapid programming etc.

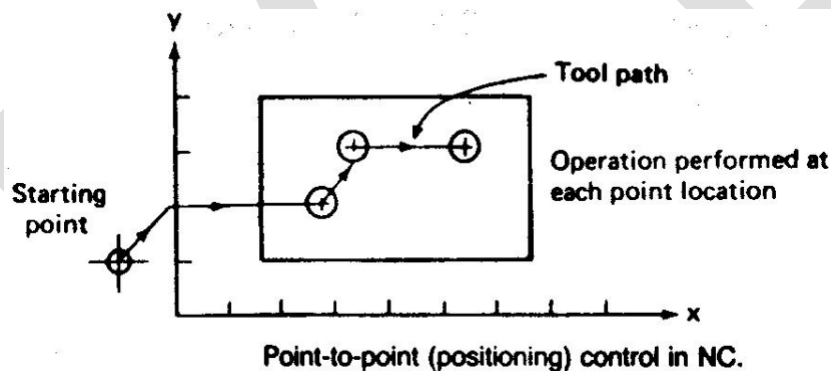
Limitations of NC

1. High initial cost of equipment
2. Need for part programming
3. Special maintenance requirements
4. More costly breakdowns

MOTION CONTROL SYSTEMS IN NC

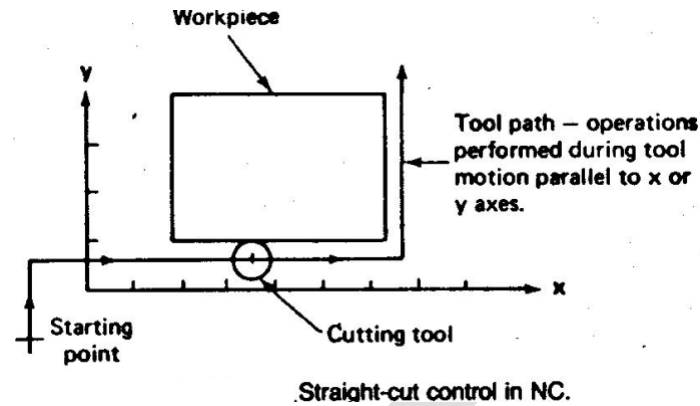
1. Point To Point (PTP) system

- The objective of this system is to move the cutting tool to a predefined location.
- The speed or path of this movement is not important.
- Once the tool reaches the desired location ,machining operation performed at that location.
- NC drill press is the good example of PTP system.
- Also used in reaming, boring, tapping, and punching are the examples for PTP system
- PTP systems are the simplest machine tool control system and are least expensive of the 3 types.
- PTP system is widely used in drilling and spot welding applications.



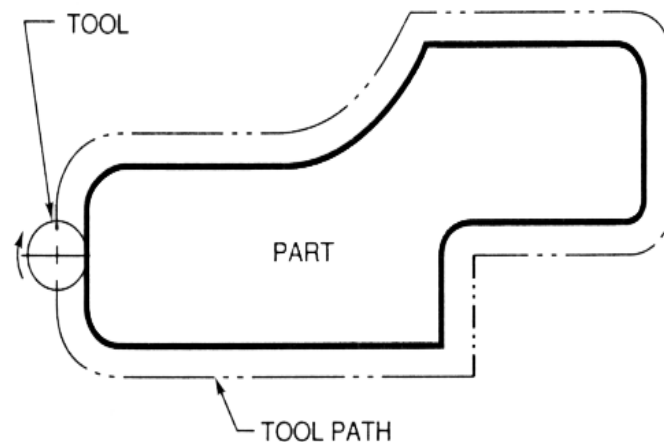
2. Straight cut NC

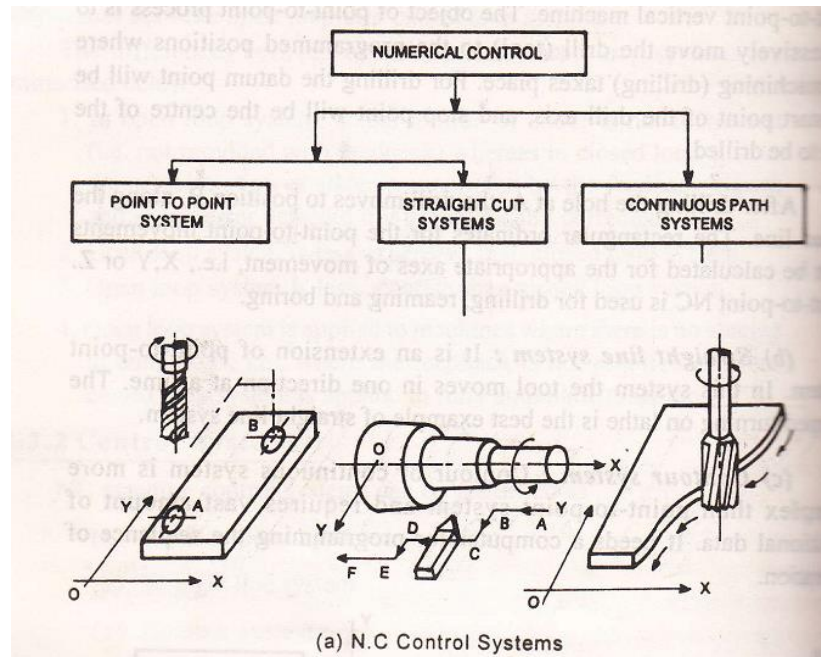
- These systems are capable of moving the cutting tool parallel to one of the major axes at a controlled rate suitable for machining.
- Suitable for performing milling operation to fabricate work piece of rectangular configurations.
- By this system it is not possible to combine the movements in more than single axis direction. Therefore angular cuts on work piece are not possible.
- An NC machine capable of straight cut movements is also capable of PTP movements



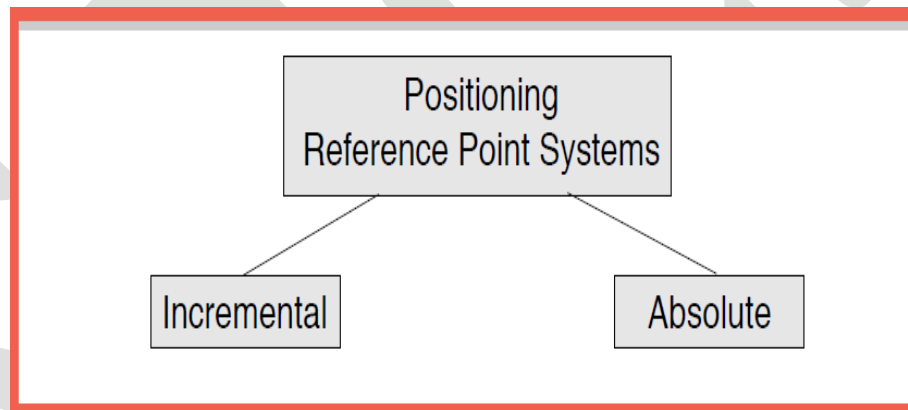
3. Contouring NC

- Contouring is the most complex, flexible and expensive type of machine tool control.
- It is capable of performing both PTP and straight cut movements.
- The specialty is simultaneous control of more than one axis of movement of the machine tool.
- The path of the cutter is continuously controlled to generate the desired geometry of the work piece. Therefore it is called continuous path system.



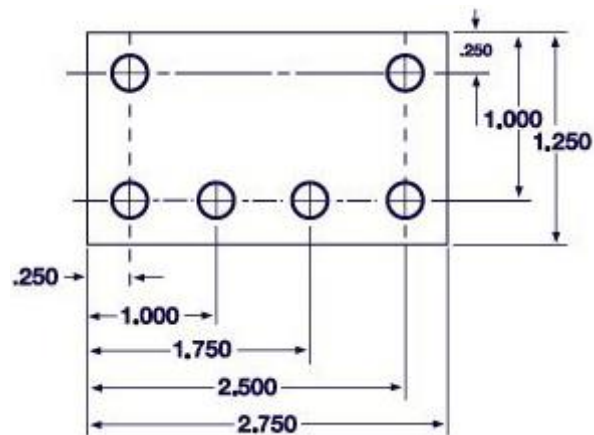


Positioning Systems in NC



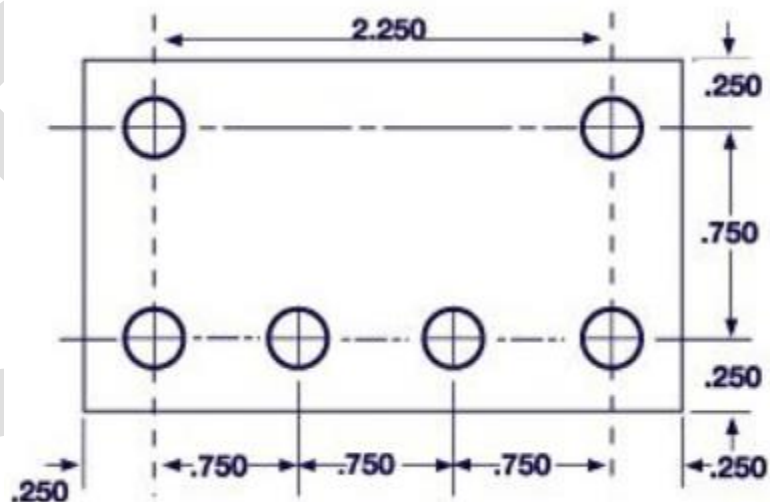
Absolute system

- Always given from a single fixed zero or origin point
- The zero or origin point may be a position on the machine table, such as the corner of the worktable or at any specific point on the work piece
- In absolute dimensioning and programming, each point or location on the work piece is given as a certain distance from the zero or reference point.



Incremental System

- Incremental program locations are always given the distance and direction from the immediately preceding point.
- Drawback is if an error occurred it will be accumulated.

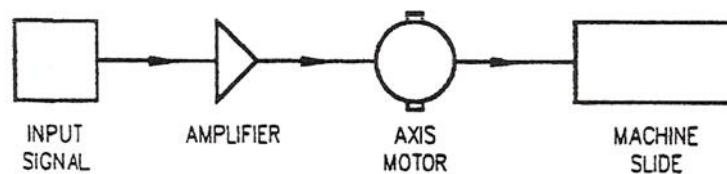


Comparison between absolute & Incremental System

<u>Absolute co-ordinate system</u>	<u>Incremental co-ordinate system</u>
In this system all the co-ordinates of point is measure from origin/datum.	In this system all the co-ordinates of point is measure from previous point.
Checking of the programme is very easy in absolute system.	Checking of the programme is very difficult in incremental system.
If mistake is made in any value of dimension in particular block, it will affect that dimension only. There is no effect on remaining program.	If mistake is made in any value of dimension in particular block, it will affect that entire program.
G90 code is used for the Absolute co-ordinate system.	G91 code is used for the incremental co-ordinate system.
It is always preferred to write main program in absolute system	It is always preferred to write subroutines in Incremental system.

Open Loop and Closed Loop Control

Open Loop – Control signals are given to actuators by MCU but movements and final destinations of the positioning system are not checked for accuracy.



Closed Loop – Equipment with transducers and sensors to measure positions , compare with control signals and correct positions as necessary.

